## Type MS · Find and fix errors in an algorithm

### Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part b.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a) | **The only correct answer is D**<br><br>*A is not correct because it is not a REPEAT UNTIL / DO WHILE loop*<br>*B is not correct because it is not a FOR loop*<br>*C is not correct because there is no such loop* | | 1 |
| 6(b)(i) | Award **one** mark for any of the following:<br><br>• `WHILE NOT (index >= LENGTH(arrival))`<br>• `WHILE NOT (index > LENGTH(arrival)-1)`<br>• `WHILE index < LENGTH(arrival)`<br>• `WHILE index <= LENGTH(arrival)-1` | Allow hard coded values for the number of items in the list.<br>For example,<br>`index >= 7`<br>`index > 6` | 1 |
| 6(b)(ii) | Award **one** mark for any of the following:<br><br>• `IF arrival[index] > 0`<br>• `IF arrival[index] >= 1` | | 1 |
| 6(b)(iii) | Award **one** mark for:<br><br>• `late` is incremented (1)<br>`SET late TO late + 1` | | 1 |

## Type MS · Find and fix errors in an algorithm

### Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part b.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(b)(iv) | Award **one** mark for:<br><br>• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)<br>`LENGTH(arrival) - (late + early)` | Allow number of trains to be hard coded as 7 in place of `LENGTH(arrival)`<br><br>Allow equivalent expressions:<br>`LENGTH(arrival) - late - early` | 1 |
| 6(c) | Award **one** mark for any of the following:<br><br>• Otherwise there would be an infinite loop (1)<br>• It is used to index all elements in the (`arrival`) array (1)<br>• To access the next value in the (`arrival`) array on the next iteration (1)<br>• So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1) | | 1 |



Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

## Type MS · Find and fix errors in an algorithm

### Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part b.iii in this mark scheme extract

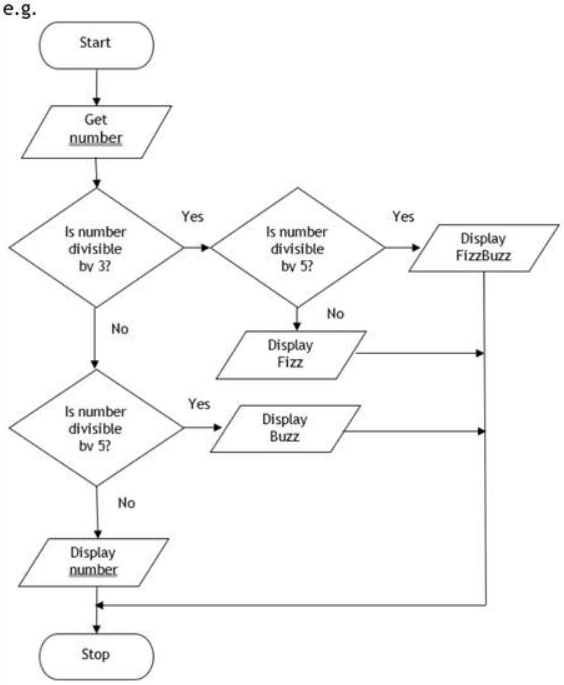
| | | | |
|---|---|---|---|
| 6(d) | Award **one** mark for each up to a maximum **six** marks:<br><br>• Get number immediately follows start (1)<br>• Number divisible by 3 and number divisible by 5 → Display Fizz (1)<br>• Number divisible by 3 and number divisible by 5 → Display Buzz (1)<br>• Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1)<br>• Number not divisible by 3 and number not divisible by 5 → Display n (1)<br>• The flowchart represents a fully functional solution (1) | For last mark point:<br>All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs<br><br>For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome.<br><br>DO NOT credit a point if additional symbols have been added. | |

## Type MS · Find and fix errors in an algorithm

### Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part b.iii in this mark scheme extract

e.g.




| | | | |
|---|---|---|---|
| | | | 6 |
| | | **Total for question 6** | **12** |

## Type MS · Find and fix errors in an algorithm

### Answer 1 of 12 · Question 6 part b.iii · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part b.iii in this mark scheme extract

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Type MS · Find and fix errors in an algorithm

Answer 2 of 12 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

| | | |
|---|---|---|
| 5(a)(i) | One mark for each correct row<br><br>**Width** / **Length** / **Height** / **Output**<br>4 / 4 / 2 / Requires large container (1)<br>2 / 2 / 2 / Too small for cargo container (1)<br>3 / 8 / 5 / Too large for cargo container (1)<br><br>Do not penalise spelling | 3 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(a)(ii) | Any response that indicates the concept of:<br><br>Determining which size shipping container is needed for the indicated cargo. | | 1 |

## Type MS · Find and fix errors in an algorithm

### Answer 2 of 12 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b) | SEND states [numStates] TO DISPLAY (1)<br>or<br>SEND state TO DISPLAY (1)<br><br>SET numStates TO numStates + 1 (1)<br>or<br>LENGTH(states) (1)<br><br>Concatenation of message and variable uses " and &<br><br>& numStates & TO DISPLAY (1)<br>or<br>& LENGTH(states) & TO DISPLAY (1)<br>or<br>STRING(numStates) TO DISPLAY (1)<br><br>Pseudocode is correct (1) | • Accept , or + in place of &<br>• Ignore capitalisation of commands<br>• | 4 |

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & numStates & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
END FOREACH
SEND "There are " & LENGTH (states) & "states" TO DISPLAY
```

```
SET numStates TO 0
SET states TO ["France", "Singapore",
               "Malta", "Panama",
               "Greece", "Italy"]
FOR EACH state FROM states DO
    SEND state TO DISPLAY
    SET numStates TO numStates + 1
END FOREACH
SEND "There are " & STRING (numStates) TO DISPLAY
```

## Type MS · Find and fix errors in an algorithm

### Answer 2 of 12 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(c)(i) | Showing that index goes up to 4 and does not exceed 4 (1)<br><br>Showing correspondence between values of index and loadWeight[index] (1) | • There does not have to be an explicit indication of the out-of-bounds / crashing error.<br>• Ignore target and found columns<br>• Allow two rows per loop with matching index and loadWeight values | 2 |

| target | found | index | loadWeight[index] |
|---|---|---|---|
| 50500 | FALSE | 0 | 20000 |
| | | 1 | 28000 |
| | | 2 | 40000 |
| | | 3 | 50000 |
| | | 4 | |

| target | found | index | loadWeight[index] |
|---|---|---|---|
| 50500 | FALSE | 0 | |
| | | 1 | 20000 |
| | | 2 | 28000 |
| | | 3 | 40000 |
| | | 4 | 50000 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(c)(ii) | 1. Use of AND operator (in the test of relevant conditions) (1)<br>2. Use of suitable function to find the length of the array using a correct comparison to index (1)<br><br>Example:<br><br>WHILE ((NOT found) AND (index < LENGTH (loadWeight))) DO | • Do not penalise pseudocode syntax<br>• Ignore changes to first condition as long as meaning is not lost | 2 |

## Type MS · Find and fix errors in an algorithm

### Answer 2 of 12 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Find and fix errors in an algorithm

Answer 3 of 12 · Question 5 part b.i · 2 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

<table>
<tr><td>5(a)(i)</td><td>Award five marks for:<br><br>• Count column correct (1)<br>• Temp column correct (1)<br>• Row 2 Numbers[Count value from row above] swapped with Temp value (1)<br>• Row 3 Numbers[Count value from row above] swapped with Temp value (1)<br>• All correct (1)<br><br>
<table>
<tr><th></th><th></th><th></th><th></th><th colspan="5">Numbers array</th></tr>
<tr><th>Count</th><th>Length</th><th>Midpoint</th><th>Temp</th><th>0</th><th>1</th><th>2</th><th>3</th><th>4</th></tr>
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<tr><td>5(a)(ii)</td><td>Award one mark from:<br><br>• To reverse the contents of the array (1)<br>• To reverse the order of the numbers (1)</td><td></td><td>1</td></tr>
<tr><td>5(a)(iii)</td><td>Award two marks for a linked explanation such as:<br><br>• You need to swap the contents of array values (1) and without Temp one of the values would be lost (1)</td><td></td><td>2</td></tr>
</table>

## Type MS · Find and fix errors in an algorithm

### Answer 3 of 12 · Question 5 part b.i · 2 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(i) | Award **two** marks for a linked explanation such as:<br>• Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1)<br>• Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1)<br>• Isaac is expecting the wrong result (1) it should be 3 (1)<br>• The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1) | | 2 |
| 5(b)(ii) | Line 3 (1)<br>Line 10 (1) | | 1 |
| 5(b)(iii) | SET count TO -1 (1)<br>SET average TO (total + 1) / (count – 1) | Ignore brackets if the meaning is clear | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(c) | Award **two** marks from:<br>• Compiler produces object code to distribute that is difficult to reverse engineer / no need to distribute the source code (1)<br>• Compiler optimises the code / object code (1)<br>• Program runs faster (as it does not need translating) (1)<br>• The target computer has no need to have the original compiler / does not need an interpreter (1)<br>• Compiled code is smaller than the original code (1), which may not fit on the DVD (1) | | 2 |

## Type MS · Find and fix errors in an algorithm

### Answer 3 of 12 · Question 5 part b.i · 2 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Find and fix errors in an algorithm

### Answer 4 of 12 · Question 5 part b.ii · 1 mark

2021 JUN · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

<table>
<tr><td>5(a)(i)</td><td>Award five marks for:<br><br>• Count column correct (1)<br>• Temp column correct (1)<br>• Row 2 Numbers[Count value from row above] swapped with Temp value (1)<br>• Row 3 Numbers[Count value from row above] swapped with Temp value (1)<br>• All correct (1)<br><br>
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## Type MS · Find and fix errors in an algorithm

### Answer 4 of 12 · Question 5 part b.ii · 1 mark

2021 JUN · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(i) | Award **two** marks for a linked explanation such as:<br>• Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1)<br>• Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1)<br>• Isaac is expecting the wrong result (1) it should be 3 (1)<br>• The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1) | | 2 |
| 5(b)(ii) | Line 3 (1)<br>Line 10 (1) | | 1 |
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### Answer 4 of 12 · Question 5 part b.ii · 1 mark

2021 JUN · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
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Type MS · Find and fix errors in an algorithm

Answer 5 of 12 · Question 5 part b.iii · 1 mark

2021 JUN · Paper 01 · extract 1 of 3 · find part b.iii in this mark scheme extract

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## Type MS · Find and fix errors in an algorithm

### Answer 5 of 12 · Question 5 part b.iii · 1 mark

2021 JUN · Paper 01 · extract 2 of 3 · find part b.iii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 5(b)(i) | Award **two** marks for a linked explanation such as:<br>• Isaac has included the -1 as a number in the addition (1) but the number has not been added to the count (1)<br>• Isaac has misunderstood the WHILE loop (1) as it should not execute after the -1 has been input (1)<br>• Isaac is expecting the wrong result (1) it should be 3 (1)<br>• The count is 1 too many as the -1 is counted as a number (1) and the total is incorrect as 1 is subtracted from the total (1) | | 2 |
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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
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## Type MS · Find and fix errors in an algorithm

### Answer 5 of 12 · Question 5 part b.iii · 1 mark

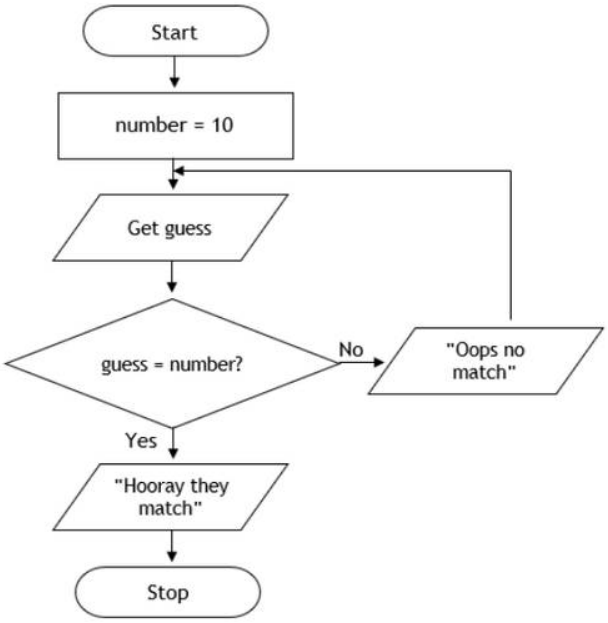
2021 JUN · Paper 01 · extract 3 of 3 · find part b.iii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Find and fix errors in an algorithm

Answer 6 of 12 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part c.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a)(i) | Award up to **five** marks for:<br><br>• Start and stop terminators in the correct positions (1) 1-3<br>• Number set to 10 AND Get guess in the correct positions (1) 4-6<br>• Loop back to before Get guess and after number set to 10 if there is no match (1)Does not need to go via message box<br>• Yes/no labels on decision match output messages (1) 4-6<br>• Correctly connected as in MS image, with at least 6 arrows correct (1) 7-9<br><br>  <br> | Boxes should be marked by content rather than shape. | 5 |
| 4(a)(ii) | **The only correct answer is C**<br><br>*A is not correct because a simulation is a completed program*<br>*B is not correct because a cipher is a form of encryption*<br>*D is not correct because a truth table is a method of testing an algorithm* | | 1 |

## Type MS · Find and fix errors in an algorithm

### Answer 6 of 12 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(b)(i) | Award **one** mark for:<br>• D | | 1 |
| 4(b)(ii) | Award **one** mark for:<br>• B | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(i) | Award up to **four** marks for:<br>• RedPoints column correct (1)<br>• OrangePoints AND NumOranges columns correct (1)<br>• Score correct in row 8 OR row 9 (1) -1 or below<br>• Outputs correct and starting on the same row as the score OR the row below (1) can all be on one line | | 4 |
| 4(c)(ii) | Award **one** mark for:<br>• 23 | | 1 |

| Colour | Score | Red Points | Orange Points | Num Oranges | Outputs |
|---|---|---|---|---|---|
| | 0 | 0 | 0 | 0 | |
| red | | 1 | | | |
| orange | | | 8 | 1 | |
| red | | 2 | | | |
| red | | 3 | | | |
| orange | | | 16 | 2 | |
| -1 | | | | | |
| | 19 | | | | |
| | | | | | Score: 19 |
| | | | | | Number of reds: 3 |
| | | | | | Number of oranges: 16 |

Type MS · Find and fix errors in an algorithm

Answer 6 of 12 · Question 4 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part c.ii in this mark scheme extract

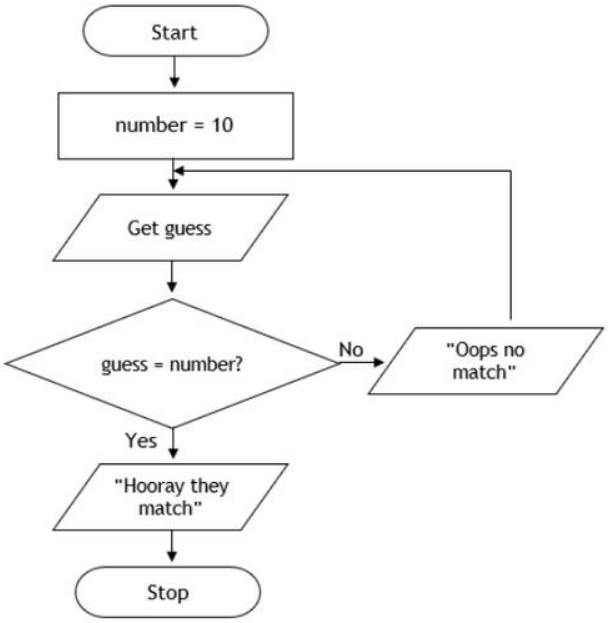
| | | | |
|---|---|---|---|
| 4(c)(iii) | Award **one** mark for:<br><br>• Pseudododode that replaces OrangePoints with NumOranges on line 23 (1)<br><br>SEND ("Number of oranges: "& NumOranges) TO DISPLAY | | 1 |
| | | **Total for question 4** | **14** |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Find and fix errors in an algorithm

Answer 7 of 12 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 01 · extract 1 of 3 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(a)(i) | Award up to **five** marks for:<br><br>• Start and stop terminators in the correct positions (1) 1-3<br>• Number set to 10 AND Get guess in the correct positions (1) 4-6<br>• Loop back to before Get guess and after number set to 10 if there is no match (1)Does not need to go via message box<br>• Yes/no labels on decision match output messages (1) 4-6<br>• Correctly connected as in MS image, with at least 6 arrows correct (1) 7-9<br><br>  <br> | Boxes should be marked by content rather than shape. | 5 |
| 4(a)(ii) | **The only correct answer is C**<br><br>*A is not correct because a simulation is a completed program*<br>*B is not correct because a cipher is a form of encryption*<br>*D is not correct because a truth table is a method of testing an algorithm* | | 1 |

Type MS · Find and fix errors in an algorithm

Answer 7 of 12 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 01 · extract 2 of 3 · find part c.iii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(b)(i) | Award **one** mark for:<br>• D | | 1 |
| 4(b)(ii) | Award **one** mark for:<br>• B | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 4(c)(i) | Award up to **four** marks for:<br>• RedPoints column correct (1)<br>• OrangePoints AND NumOranges columns correct (1)<br>• Score correct in row 8 OR row 9 (1) -1 or below<br>• Outputs correct and starting on the same row as the score OR the row below (1) can all be on one line | | 4 |
| 4(c)(ii) | Award **one** mark for:<br>• 23 | | 1 |

| Colour | Score | Red Points | Orange Points | Num Oranges | Outputs |
|---|---|---|---|---|---|
| | 0 | 0 | 0 | 0 | |
| red | | 1 | | | |
| orange | | | 8 | 1 | |
| red | | 2 | | | |
| red | | 3 | | | |
| orange | | | 16 | 2 | |
| -1 | | | | | |
| | 19 | | | | |
| | | | | | Score: 19 |
| | | | | | Number of reds: 3 |
| | | | | | Number of oranges: 16 |

Type MS · Find and fix errors in an algorithm

Answer 7 of 12 · Question 4 part c.iii · 1 mark

2022 JUN · Paper 01 · extract 3 of 3 · find part c.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 4(c)(iii) | Award **one** mark for:<br><br>• Pseudododode that replaces OrangePoints with NumOranges on line 23 (1)<br><br>SEND ("Number of oranges: "& NumOranges) TO DISPLAY | | 1 |
| | | **Total for question 4** | **14** |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Find and fix errors in an algorithm

### Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 1 of 9 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 5(a) | A1<br>A2 | Award 1 mark for demonstration of one correctly merged/split row<br>Award 2 marks for correct merge sort<br><br>Blackfin, Bigeye \| Longtail, Albacore \| Bluefin<br><br>Longtail, Blackfin, Bigeye, Albacore \| Bluefin<br><br>Longtail, Bluefin, Blackfin, Bigeye, Albacore<br><br>OR<br><br>Bigeye \| Blackfin, Albacore \| Longtail, Bluefin<br><br>Bigeye \| Longtail, Bluefin, Blackfin, Albacore<br><br>Longtail, Bluefin, Blackfin, Bigeye, Albacore<br><br>OR<br><br>Blackfin, Bigeye \| Albacore \| Longtail, Bluefin<br><br>Blackfin, Bigeye \| Longtail, Bluefin, Albacore | We are testing pupils understanding of how a merge sort progresses:<br>• Divide and conquer (split)<br>• Merging adjacent elements<br>Allow answers in ascending order | 2 |

## Type MS · Find and fix errors in an algorithm

### Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 2 of 9 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| | | Longtail, Bluefin, Blackfin, Bigeye, Albacore<br><br>OR<br><br>Blackfin, Bigeye \| Albacore \| Longtail, Bluefin<br><br>Blackfin, Bigeye, Albacore \| Longtail, Bluefin<br><br>Longtail, Bluefin, Blackfin, Bigeye, Albacore | | |

## Type MS · Find and fix errors in an algorithm

### Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 3 of 9 · find part b in this mark scheme extract

<table>
<tr><th>Question</th><th>mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(b)</td><td>B1<br>B2</td><td>Award one mark for each correct cell:<br><table><tr><td>Line number with error</td><td>11 (1)</td></tr><tr><td>Corrected line of pseudocode</td><td>SET tmp TO myTuna[ndx] (1)</td></tr></table></td><td>Ignore spellings<br><br>Ignore missing text as long as the minimum of [ndx] is provided<br><br>B2 does not depend on B1</td><td>2</td></tr>
</table>

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 5(c) | Award **one** mark for each of: | | C1 allow append as writing to file<br><br>C4 Do not award comma within an array<br><br>C5 must have 1 complete line written to file (allow additional spaces and commas). Should include individual elements (2D index) and not a 1D array from tbl_tuna<br><br>C6 complete file must be as expected e.g. no additional comma at end of line and no spaces around commas | **6** |

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 4 of 9 · find part b in this mark scheme extract

**C#**

```
// Q05cFINISHED

using System.IO;

String[,] tblTuna =  {
    { "Yellowfin", "105", "15", "3"},
    { "Albacore","90","15","5"},
    { "Skipjack","50","3","4"},
    { "Bigeye","105","25","4"},
    { "Atlantic Bonito","50","4","2"},
    { "Northern Bluefin","190","120","11"},
    { "Southern Bluefin","190","120","11"},
    { "Tongol","90","20","4"}
};



// ------------------------------------------
// Write your code below this line

String fileName = "TunaData.txt";       // Output file name
String comma = ",";                     // Use as a constant
String full_path = "C:\\Q05c";
fileName = full_path + "\\" + fileName;

// Create variables as needed
int number = 101;           // Number for the line at the front
String lineOut = "";        // The line to output

// Get the dimensions of the array for looping
int rows = tblTuna.GetLength(0);
int columns = tblTuna.GetLength(1);

// Open the file for writing
StreamWriter fileWriter = new StreamWriter(fileName);

```

## Type MS · Find and fix errors in an algorithm

### Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 5 of 9 · find part b in this mark scheme extract

```
37  // Process every tuna in the table
38  for (int i = 0; i < rows; i++)          // Rows
39  {
40      // Start with the number and a comma
41      lineOut = number.ToString() + comma;
42
43      for (int j = 0; j < columns; j++)   // Columns
44      {
45          // Add the name and the numbers
46          lineOut = lineOut + tblTuna[i, j];
47
48          // Add a comma to all the fields except the last
49          if (j < columns - 1)
50          {
51              lineOut = lineOut + comma;
52          }
53      }
54      // Write the line to the file
55      fileWriter.WriteLine(lineOut);
56
57      // Go to the next number for the line
58      number = number + 1;
59  }
60
61  // Close the file
62  fileWriter.Close();
```

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 6 of 9 · find part b in this mark scheme extract

**Java**

```java
// Q05cFINISHED

import java.io.FileWriter;
import java.io.IOException;

public class Q05cFINISHED {

    public static void main(String[] args) throws Exception {

        String[][] tblTuna =
        {
            { "Yellowfin", "105", "15", "3"},
            { "Albacore","90","15","5"},
            { "Skipjack","50","3","4"},
            { "Bigeye","105","25","4"},
            { "Atlantic Bonito","50","4","2"},
            { "Northern Bluefin","190","120","11"},
            { "Southern Bluefin","190","120","11"},
            { "Tongol","90","20","4"}
        };

        // ------------------------------------------
        // Write your code below this
        String fileName = "TunaData.txt";       // Output file name
        String comma = ",";                     // Use as a constant
        String full_path = "C:\\src\\q05c";
        fileName = full_path + "\\" + fileName;

        // Create variables as needed
        int number = 101;           // Number for the line at the front
        String lineOut = "";        // The line to output
        String linefeed = "\n";     // Use as a constant

        // Get the dimensions of the array for looping
        int rows = tblTuna.length;
        int columns = tblTuna[0].length;

        // Open the file for writing
        FileWriter fileWriter = new FileWriter(fileName);

```

## Type MS · Find and fix errors in an algorithm

### Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 7 of 9 · find part b in this mark scheme extract

```
41          // Process every tuna in the table
42          for (int i = 0; i < rows; i++)          // Rows
43          {
44              // Start with the number and a comma
45              lineOut = Integer.toString(number) + comma;
46
47              for (int j = 0; j < columns; j++)   // Columns
48              {
49                  // Add the name and the numbers
50                  lineOut = lineOut + tblTuna[i][j];
51
52                  // Add a comma to all the fields except the last
53                  if (j < columns - 1)
54                  {
55                      lineOut = lineOut + comma;
56                  }
57                  else
58                  {
59                      lineOut = lineOut + linefeed;
60                  }
61              }
62              // Write the line to the file
63              fileWriter.write(lineOut);
64
65              // Go to the next number for the line
66              number = number + 1;
67          }
68
69          // Close the file
70          fileWriter.close();
71      }
72
73  }
```

Type MS · Find and fix errors in an algorithm

Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 8 of 9 · find part b in this mark scheme extract

**Python**

```
# Q05cFINISHED

tbl_tuna = [["Yellowfin",105,15,3],
            ["Albacore",90,15,5],
            ["Skipjack",50,3,4],
            ["Bigeye",105,25,4],
            ["Atlantic Bonito",50,4,2],
            ["Northern Bluefin",190,120,11],
            ["Southern Bluefin",190,120,11],
            ["Tongol",90,20,4]]

# ---------------------------------------------
# Write your code below this line

FILE_OUT = "TunaData.txt"           # Output file name
COMMA = ","                         # Use as constant

# Create variables as needed
LINEFEED = "\n"
number = 101                        # Number at the front
line_out = ""                       # The line to write

# Open the file for writing
file = open (FILE_OUT, "w")

# Process every tuna in the table
for tuna in tbl_tuna:
    # Start with the number and a comma
    line_out = str (number) + COMMA

    # Add the name of the tuna and a comma
    line_out = line_out + tuna[0] + COMMA

    # Each of the numbers need to be converted to a string
    # before adding to the output string.
    line_out = line_out + str (tuna[1]) + COMMA
    line_out = line_out + str (tuna[2]) + COMMA
    line_out = line_out + str (tuna[3])     # No comma on last field

    # Add a line feed to the whole line
    line_out = line_out + LINEFEED

    # Write the line to the file
    file.write (line_out)

    # Go to the next number for the line
    number = number + 1

# Close the file
file.close ()
```

## Type MS · Find and fix errors in an algorithm

### Answer 8 of 12 · Question 5 part b · 2 marks

2023 JUN · Paper 02 · extract 9 of 9 · find part b in this mark scheme extract

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|

## Type MS · Find and fix errors in an algorithm

### Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part b.i in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 6(a) | **The only correct answer is D**<br><br>*A is not correct because it is not a REPEAT UNTIL / DO WHILE loop*<br>*B is not correct because it is not a FOR loop*<br>*C is not correct because there is no such loop* | | 1 |
| 6(b)(i) | Award **one** mark for any of the following:<br><br>• `WHILE NOT (index >= LENGTH(arrival))`<br>• `WHILE NOT (index > LENGTH(arrival)-1)`<br>• `WHILE index < LENGTH(arrival)`<br>• `WHILE index <= LENGTH(arrival)-1` | Allow hard coded values for the number of items in the list.<br>For example,<br>`index >= 7`<br>`index > 6` | 1 |
| 6(b)(ii) | Award **one** mark for any of the following:<br><br>• `IF arrival[index] > 0`<br>• `IF arrival[index] >= 1` | | 1 |
| 6(b)(iii) | Award **one** mark for:<br><br>• `late` is incremented (1)<br>`SET late TO late + 1` | | 1 |

## Type MS · Find and fix errors in an algorithm

### Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(b)(iv) | Award **one** mark for:<br><br>• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)<br>`LENGTH(arrival) - (late + early)` | Allow number of trains to be hard coded as 7 in place of<br>`LENGTH(arrival)`<br><br>Allow equivalent expressions:<br>`LENGTH(arrival) - late - early` | 1 |
| 6(c) | Award **one** mark for any of the following:<br><br>• Otherwise there would be an infinite loop (1)<br>• It is used to index all elements in the (`arrival`) array (1)<br>• To access the next value in the (`arrival`) array on the next iteration (1)<br>• So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1) | | 1 |

## Type MS · Find and fix errors in an algorithm

### Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part b.i in this mark scheme extract

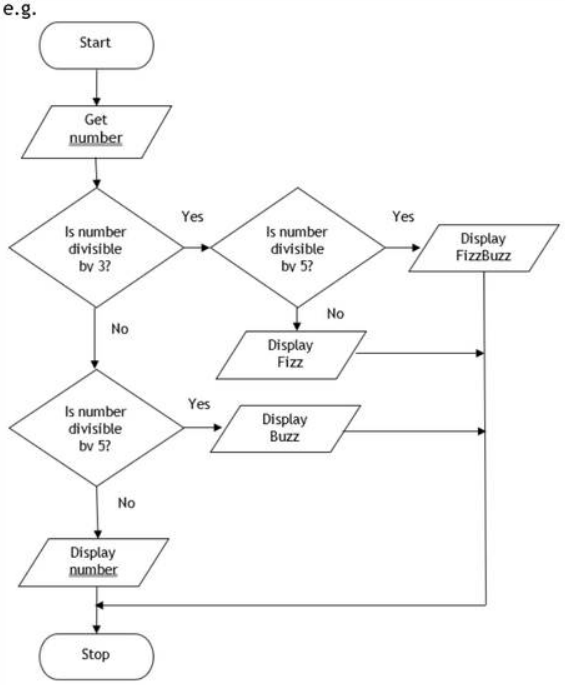
| | | | |
|---|---|---|---|
| 6(d) | Award **one** mark for each up to a maximum **six** marks:<br><br>• Get number immediately follows start (1)<br>• Number divisible by 3 and number divisible by 5 → Display Fizz (1)<br>• Number divisible by 3 and number divisible by 5 → Display Buzz (1)<br>• Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1)<br>• Number not divisible by 3 and number not divisible by 5 → Display n (1)<br>• The flowchart represents a fully functional solution (1) | For last mark point:<br>All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs<br><br>For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome.<br><br>DO NOT credit a point if additional symbols have been added. | |

## Type MS · Find and fix errors in an algorithm

### Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part b.i in this mark scheme extract

e.g.




| | | | |
|---|---|---|---|
| | | | 6 |
| | | **Total for question 6** | **12** |

22

## Type MS · Find and fix errors in an algorithm

### Answer 9 of 12 · Question 6 part b.i · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part b.i in this mark scheme extract

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## Type MS · Find and fix errors in an algorithm

### Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a) | **The only correct answer is D**<br><br>*A is not correct because it is not a REPEAT UNTIL / DO WHILE loop*<br>*B is not correct because it is not a FOR loop*<br>*C is not correct because there is no such loop* | | 1 |
| 6(b)(i) | Award **one** mark for any of the following:<br><br>• `WHILE NOT (index >= LENGTH(arrival))`<br>• `WHILE NOT (index > LENGTH(arrival)-1)`<br>• `WHILE index < LENGTH(arrival)`<br>• `WHILE index <= LENGTH(arrival)-1` | Allow hard coded values for the number of items in the list.<br>For example,<br>`index >= 7`<br>`index > 6` | 1 |
| 6(b)(ii) | Award **one** mark for any of the following:<br><br>• `IF arrival[index] > 0`<br>• `IF arrival[index] >= 1` | | 1 |
| 6(b)(iii) | Award **one** mark for:<br><br>• `late` is incremented (1)<br>`SET late TO late + 1` | | 1 |

## Type MS · Find and fix errors in an algorithm

### Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(b)(iv) | Award **one** mark for:<br><br>• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)<br>`LENGTH(arrival) - (late + early)` | Allow number of trains to be hard coded as 7 in place of<br>`LENGTH(arrival)`<br><br>Allow equivalent expressions:<br>`LENGTH(arrival) - late - early` | 1 |
| 6(c) | Award **one** mark for any of the following:<br><br>• Otherwise there would be an infinite loop (1)<br>• It is used to index all elements in the (`arrival`) array (1)<br>• To access the next value in the (`arrival`) array on the next iteration (1)<br>• So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1) | | 1 |

## Type MS · Find and fix errors in an algorithm

### Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part b.ii in this mark scheme extract

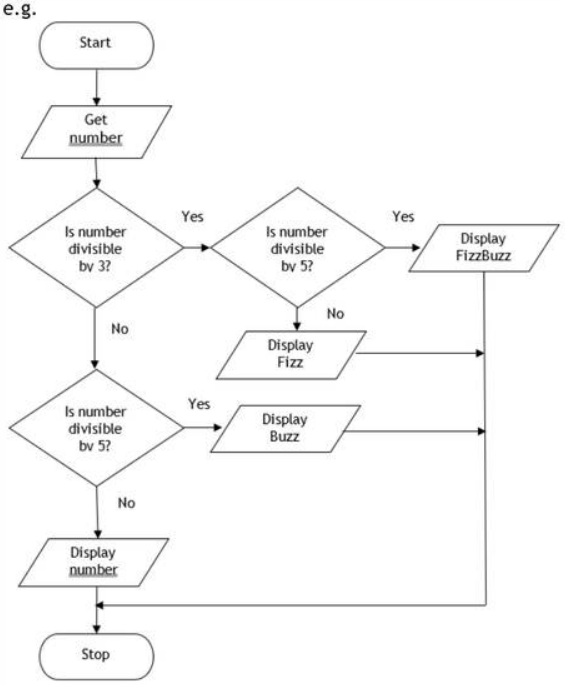
| | | | |
|---|---|---|---|
| 6(d) | Award **one** mark for each up to a maximum **six** marks:<br><br>• Get number immediately follows start (1)<br>• Number divisible by 3 and number divisible by 5 → Display Fizz (1)<br>• Number divisible by 3 and number divisible by 5 → Display Buzz (1)<br>• Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1)<br>• Number not divisible by 3 and number not divisible by 5 → Display n (1)<br>• The flowchart represents a fully functional solution (1) | For last mark point:<br>All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs<br><br>For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome.<br><br>DO NOT credit a point if additional symbols have been added. | |

## Type MS · Find and fix errors in an algorithm

### Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part b.ii in this mark scheme extract

e.g.




| | | | |
|---|---|---|---|
| | | | 6 |
| | | **Total for question 6** | **12** |

## Type MS · Find and fix errors in an algorithm

### Answer 10 of 12 · Question 6 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part b.ii in this mark scheme extract

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## Type MS · Find and fix errors in an algorithm

### Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part b.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(a) | **The only correct answer is D**<br><br>*A is not correct because it is not a REPEAT UNTIL / DO WHILE loop*<br>*B is not correct because it is not a FOR loop*<br>*C is not correct because there is no such loop* | | 1 |
| 6(b)(i) | Award **one** mark for any of the following:<br><br>• `WHILE NOT (index >= LENGTH(arrival))`<br>• `WHILE NOT (index > LENGTH(arrival)-1)`<br>• `WHILE index < LENGTH(arrival)`<br>• `WHILE index <= LENGTH(arrival)-1` | Allow hard coded values for the number of items in the list.<br>For example,<br>`index >= 7`<br>`index > 6` | 1 |
| 6(b)(ii) | Award **one** mark for any of the following:<br><br>• `IF arrival[index] > 0`<br>• `IF arrival[index] >= 1` | | 1 |
| 6(b)(iii) | Award **one** mark for:<br><br>• `late` is incremented (1)<br>`SET late TO late + 1` | | 1 |

## Type MS · Find and fix errors in an algorithm

### Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part b.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 6(b)(iv) | Award **one** mark for:<br><br>• On time is calculated as the remainder when early and late trains are subtracted from the total number of trains (1)<br>`LENGTH(arrival) - (late + early)` | Allow number of trains to be hard coded as 7 in place of<br>`LENGTH(arrival)`<br><br>Allow equivalent expressions:<br>`LENGTH(arrival) - late - early` | 1 |
| 6(c) | Award **one** mark for any of the following:<br><br>• Otherwise there would be an infinite loop (1)<br>• It is used to index all elements in the (`arrival`) array (1)<br>• To access the next value in the (`arrival`) array on the next iteration (1)<br>• So that the condition used in the while loop evaluates to True when all the elements in arrival have been processed (1) | | 1 |



Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

## Type MS · Find and fix errors in an algorithm

### Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part b.iv in this mark scheme extract

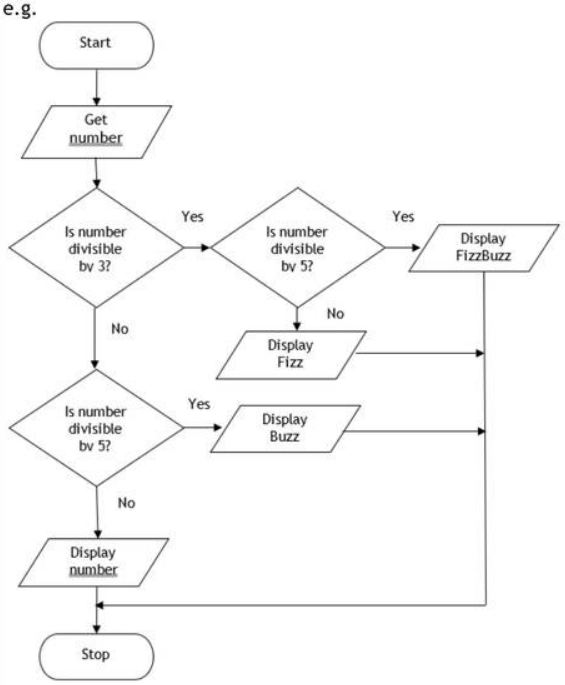
| | | | |
|---|---|---|---|
| 6(d) | Award **one** mark for each up to a maximum **six** marks:<br><br>• Get number immediately follows start (1)<br>• Number divisible by 3 and number divisible by 5 → Display Fizz (1)<br>• Number divisible by 3 and number divisible by 5 → Display Buzz (1)<br>• Number divisible by 3 and number divisible by 5 → Display FizzBuzz (1)<br>• Number not divisible by 3 and number not divisible by 5 → Display n (1)<br>• The flowchart represents a fully functional solution (1) | For last mark point:<br>All symbols fully connected with directed arrows. Start and stop terminators used correctly. No hanging symbols. Decision boxes have exactly two labelled outputs<br><br>For each of the first five mark points the mark should only be awarded if there is one correct logical path to get to the outcome.<br><br>DO NOT credit a point if additional symbols have been added. | |

## Type MS · Find and fix errors in an algorithm

### Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part b.iv in this mark scheme extract

e.g.




| | | | |
|---|---|---|---|
| | | | 6 |
| | | **Total for question 6** | **12** |

## Type MS · Find and fix errors in an algorithm

### Answer 11 of 12 · Question 6 part b.iv · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part b.iv in this mark scheme extract

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## Type MS · Find and fix errors in an algorithm

### Answer 12 of 12 · Question 5 part a.i · 2 marks

2025 JUN · Paper 02 · extract 1 of 4 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 5(a)(i) | A1<br>A2 | final score not accurate (1) because score is resetting/score keeps going back to zero (in each iteration of the loop) (1) | 'Score not accurate' is not enough on its own<br>'Score on line 20' or 'Score output to DISPLAY' would count as final score. | **2** |

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| | | | | |
| 5(a)(ii) | A3<br>A4 | Award up to **two** marks for a linked description such as:<br>• Asking 10 addition questions (1) and calculating final score (1)<br>• Asking randomised adding questions (1) and checking user answer against computer answer (1)<br>• Creating adding quiz with random numbers up to 99 (1) and outputting number of correct answers (1) | Processes that can be included<br>Concept of this being an 'Adding Quiz' required<br>• Getting random numbers for question<br>• Calculating sum of two random numbers for correct answers<br>• comparing user answer to correct answer<br>• incrementing /tracking score<br>• asking 10 questions (loop)<br>• outputting final score | **2** |

## Type MS · Find and fix errors in an algorithm

### Answer 12 of 12 · Question 5 part a.i · 2 marks

2025 JUN · Paper 02 · extract 2 of 4 · find part a.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|
| 5(b) | | | | |
| | B1 | Open file for writing (1) | "w" mode only<br>Accept "w+" | |
| | B2 | Selection used to get grade (1) | Must address all 4 grades<br>Does not have to be accurate on boundaries e.g. >80 is ok for distinction etc.1 | |
| | B3 | Output suitable message to say file created (1) | | |
| | B4 | Evidence of 2D indexing (1) | Even if element not used in rest of program<br>2D indexing **or** 1D index within *for-each* structure **or** sum(1Darray) or min(1Darray) etc.<br>At least 1 inner value accessed | |
| | B5 | Lowest score **must** be found and omitted/subtracted from calculated scores (all arrays) (1) | Any method e.g.:<br>• List methods<br>• Selection statements<br>• iterable functions<br>must be done programmatically | |
| | B6 | At least 1 line written to file (1) | Must contain something generated by program (total/result), does not have to be correct total or grade | |
| | B7 | Add line feed to write line after both score and grade (1) | | |
| | B8 | Output file all correct (1) | | **8** |

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# Type MS · Find and fix errors in an algorithm

## Answer 12 of 12 · Question 5 part a.i · 2 marks

2025 JUN · Paper 02 · extract 3 of 4 · find part a.i in this mark scheme extract



```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
total = 0
grade = ""
outString = ""

with open ("grades.txt","w") as theFile:

    # calculate each students score
    for student in Tbl_scores:
        student.sort()
        total = student[1] + student[2]

        # decide student grade
        if total >= 80:
            grade = "Distinction"
        elif total >= 65:
            grade = "Merit"
        elif total >= 50:
            grade = "Pass"
        else:
            grade = "Fail"

        # write score and grade to file
        outString = str(total) + " " + grade + "\n"
        theFile.write(outString)

print ("Grade file has been created")
```

```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
total = 0
grade = ""
outString = ""

theFile = open ("grades.txt","w")

# calculate each students score
for student in Tbl_scores:
    total = sum(student) - min(student)

    # decide student grade
    if total >= 80:
        grade = "Distinction"
    elif total >= 65:
        grade = "Merit"
    elif total >= 50:
        grade = "Pass"
    else:
        grade = "Fail"

    # write score and grade to file
    outString = str(total) + " " + grade + "\n"
    theFile.write(outString)

theFile.close()

print ("Grade file has been created")
```

```
# Q05b

Tbl_scores = [[45,42,43],[19,36,41],[48,48,48],[35,32,31],
              [18,22,26],[41,35,31],[15,19,12],[41,39,40],
              [27,25,22],[32,36,35],[37,36,37],[40,35,41]]

# ----------------------------------------------
# Write your code below this line
grade = ""
outString = ""

theFile = open ("grades.txt","w")

# calculate each students score
for student in Tbl_scores:
    total = 0
    lowest = 51
    for score in student:
        if score < lowest:
            lowest = score
        total = total + score
    total = total - lowest

    # decide student grade
    if total >= 80:
        grade = "Distinction"
    elif total >= 65:
        grade = "Merit"
    elif total >= 50:
        grade = "Pass"
    else:
        grade = "Fail"

    # write score and grade to file
    outString = str(total) + " " + grade + "\n"
    theFile.write(outString)

theFile.close()

print ("Grade file has been created")
```

## Type MS · Find and fix errors in an algorithm

### Answer 12 of 12 · Question 5 part a.i · 2 marks

2025 JUN · Paper 02 · extract 4 of 4 · find part a.i in this mark scheme extract

Paper 2 – Application of Computational Thinking

| Question | mp | Answer | Additional guidance | Mark |
|---|---|---|---|---|